# CP Violation and the Strong CP Problem from Recognition Science: $\theta_{\mathrm{QCD}} = 0$ from $J$-Symmetry

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**Abstract**

We derive the complete CP structure of the Standard Model from Recognition Science (RS). The strong CP angle is exactly zero: $\theta_{\mathrm{eff}} = 0$ from the $J$-symmetry $J(x) = J(x^{-1})$, without requiring an axion or Peccei–Quinn mechanism. This resolves the strong CP problem because both $\theta_{\mathrm{QCD}} = 0$ (from the CP-invariance of the $J$-cost action) and $\mathrm{argdet}(M_u M_d) = 0$ (from the reality of $\varphi$-ladder mass eigenvalues) are structurally forced. CP violation in the weak sector arises from the three-generation torsion $\tau_g \in \{0, 11, 17\}$: the non-commutativity of the generation mixing produces a physical CKM phase $\delta_{\mathrm{KM}}$. CPT invariance is exact, forced by the double-entry ledger structure. The sharp falsifier is: detection of a QCD axion coupling to $G\tilde{G}$ would require revision of RS.

## 1 Introduction: The Two CP Problems

CP violation in the Standard Model comes in two forms:

1. **Weak CP violation**: The CKM matrix has one irreducible complex phase $\delta_{\mathrm{KM}}$, experimentally confirmed in $K$ and $B$ meson decays [4].

2. **The strong CP problem**: The QCD Lagrangian admits a topological term

$$\mathcal{L}_\theta = \frac{\theta_{\mathrm{eff}}\, g_s^2}{32\pi^2}\, G^a_{\mu\nu} \tilde{G}^{a\mu\nu}, \tag{1}$$

   where $\theta_{\mathrm{eff}} = \theta_{\mathrm{QCD}} + \mathrm{argdet}(M_u M_d)$ is the physically observable angle, incorporating both the bare QCD vacuum angle and the phase of the quark mass matrix determinant. Experimentally, $|\theta_{\mathrm{eff}}| < 10^{-10}$ from neutron EDM bounds [5], while it could *a priori* be $O(1)$.

The Standard Model offers no explanation for why $\theta_{\mathrm{eff}}$ is so small. The Peccei–Quinn (PQ) mechanism [3] introduces a new $U(1)_{\mathrm{PQ}}$ symmetry that dynamically relaxes $\theta_{\mathrm{eff}} \to 0$ via an axion field. RS provides a structural alternative: $\theta_{\mathrm{eff}} = 0$ is forced by the $J$-cost symmetry, with no additional fields or symmetries.

# 2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

**Definition 2.1** (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

**Axiom A1 (Normalization):** $J(1) = 0$.
**Axiom A2 (Recognition Composition Law, RCL):** $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
**Axiom A3 (Calibration):** $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

**Theorem 2.2** (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

*Proof sketch.* Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

The key property for CP structure is the *inversion symmetry* $J(x) = J(x^{-1})$, which follows immediately from the definition: $\frac{1}{2}(x^{-1} + x) - 1 = J(x)$. This symmetry forces the gauge action to be CP-invariant, setting $\theta_{\text{QCD}} = 0$.

# 3 Strong CP: $\theta_{\text{eff}} = 0$ from $J$-Symmetry

## 3.1 The $J$-Symmetry

**Theorem 3.1** ($J$-Inversion Symmetry)**.** *The RS cost functional satisfies exact inversion symmetry:*

$$J(x) = J(x^{-1}) \qquad \forall\, x > 0. \tag{2}$$

*Proof.* From $J(x) = \frac{1}{2}(x + x^{-1}) - 1$:

$$J(x^{-1}) = \tfrac{1}{2}(x^{-1} + x) - 1 = J(x). \tag{3}$$

□

## 3.2 $\theta_{\text{QCD}} = 0$ from $J$-Invariance of the Action

**Theorem 3.2** (Strong CP Vanishing: Bare Angle)**.** *In the RS framework,* $\theta_{\text{QCD}} = 0$ *exactly.*

*Proof.* The RS action for gauge fields on the ledger is constructed from $J$-costs of link variables. For a gauge link $U_\ell = e^{ig_s A_\mu \ell^\mu}$, the $J$-cost contribution is $J(U_\ell)$, which satisfies

$$J(e^{i\phi}) = J(e^{-i\phi}) \qquad \forall\, \phi \in \mathbb{R}. \tag{4}$$

The extension of $J$ to complex arguments of unit modulus is $J(e^{i\phi}) = \frac{1}{2}(e^{i\phi} + e^{-i\phi}) - 1 = \cos\phi - 1$, which is real and depends only on $|\phi|$ (even in $\phi$). This extends the original $J : (0, \infty) \to [0, \infty)$ to the unit circle by replacing $x + x^{-1}$ with $e^{i\phi} + e^{-i\phi} = 2\cos\phi$; the resulting function is the natural complex-valued extension. Under CP, the gauge link transforms as $U_\ell \to U_\ell^{-1} = e^{-ig_s A_\mu \ell^\mu}$. Therefore:

$$S_{\text{RS}}[A] = \sum_{\text{links}} J(U_\ell) = \sum_{\text{links}} J(U_\ell^{-1}) = S_{\text{RS}}[A^{\text{CP}}]. \tag{5}$$

The RS action is exactly CP-invariant for the gauge sector. Since the $\theta$ term (1) violates CP ($\mathcal{L}_\theta \to -\mathcal{L}_\theta$ under CP), no such term can appear in the RS-derived action. Therefore $\theta_{\text{QCD}} = 0$. □

### 3.3 $\text{argdet}(M_u M_d) = 0$ from $\varphi$-Ladder Reality

**Theorem 3.3** (Strong CP Vanishing: Mass Phase)**.** *In RS,* $\text{argdet}(M_u M_d) = 0$ *exactly.*

*Proof.* The RS mass law assigns each quark a position on the $\varphi$-ladder:

$$m_f = Y_s \cdot E_{\text{coh}} \cdot \varphi^{r_f - 8 + \text{gap}(Z_f) + f_f^{\text{RG}}}. \tag{6}$$

All quantities on the right-hand side are real: $Y_s \in \mathbb{Z}$, $\varphi \in \mathbb{R}^+$, $r_f \in \mathbb{Z}$, $\text{gap}(Z_f) \in \mathbb{Z}$, and $f_f^{\text{RG}} \in \mathbb{R}$. Therefore the mass eigenvalues $m_f$ are strictly real and positive.

The determinant of the quark mass matrices factorizes as:

$$\det(M_u) = m_u \cdot m_c \cdot m_t > 0, \qquad \det(M_d) = m_d \cdot m_s \cdot m_b > 0. \tag{7}$$

Since both determinants are real and positive, $\text{argdet}(M_u M_d) = 0$. □

**Corollary 3.4** (Strong CP Resolution)**.** *The physically observable strong CP angle vanishes:*

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \text{argdet}(M_u M_d) = 0 + 0 = 0. \tag{8}$$

*This is a structural identity, not a fine-tuning or dynamical relaxation.*

### 3.4 No Axion Required

**Corollary 3.5** (No QCD Axion)**.** *Since* $\theta_{\text{eff}} = 0$ *structurally (Corollary 3.4), the Peccei–Quinn mechanism [3] is unnecessary. No QCD axion coupling to* $G\tilde{G}$ *at the QCD scale exists.*

*Remark* 3.6. This does not exclude the existence of axion-like particles (ALPs) with no QCD coupling. The RS prediction is specifically about the QCD axion: the particle invented to solve the strong CP problem. Since the problem does not exist in RS, neither does its solution.

## 4 Electric Dipole Moments

The neutron electric dipole moment (EDM) from a nonzero $\theta_{\text{eff}}$ is [7]:

$$d_n \approx \theta_{\text{eff}} \cdot 3.6 \times 10^{-16}\ e \cdot \text{cm}. \tag{9}$$

**Proposition 4.1** (Neutron EDM Prediction)**.** *With* $\theta_{\text{eff}} = 0$*, the only contribution to* $d_n$ *comes from SM electroweak loops (CKM CP violation):*

$$d_n^{\text{RS}} \approx d_n^{\text{SM}} \approx 10^{-32}\ e \cdot \text{cm}. \tag{10}$$

The current experimental limit is $|d_n| < 1.8 \times 10^{-26}\ e \cdot \text{cm}$ (PSI 2020) [5]. Next-generation experiments (nEDM@SNS, TUCAN) aim for $\sim 10^{-28}\ e \cdot \text{cm}$.

*Remark* 4.2. A nonzero $d_n$ at the $10^{-28}$ level (beyond the SM prediction) would indicate new sources of CP violation not present in RS and would require revision of the theory.

# 5 CPT Invariance from the Double-Entry Ledger

**Theorem 5.1** (CPT Invariance)**.** *CPT invariance holds exactly in the RS framework, forced by the double-entry ledger structure.*

*Proof.* The proof follows from three structural properties of the ledger:

**(i) Charge conjugation (C).** The double-entry ledger requires that every recognition event has a balancing anti-event: if a charge $+q$ is recorded at voxel $\mathbf{n}$, a charge $-q$ must be recorded elsewhere. This is the ledger implementation of charge conjugation.

**(ii) Parity (P).** The $J$-cost functional depends only on the ratio $x$, not on the spatial orientation. The ledger lattice $\mathbb{Z}^3$ has the full octahedral symmetry group, which includes spatial inversion $\mathbf{r} \to -\mathbf{r}$.

**(iii) Time reversal (T).** The $J$-symmetry $J(x) = J(x^{-1})$ is the temporal analog of parity: it equates a process ($x$) with its time-reverse ($x^{-1}$). (Note: this is T-symmetry at the level of *individual recognition events*, distinct from the macroscopic arrow of time—which is broken because the ledger's tick index only advances, not because individual $J$-costs distinguish $x$ from $x^{-1}$. CPT invariance is a property of the microscopic action; the arrow of time is a property of the macroscopic dynamics. There is no contradiction.)

**(iv) CPT combined.** The combined CPT operation maps every ledger entry to its complete conjugate: charge-reversed, spatially-reflected, temporally-reversed. The double-entry structure guarantees that if a configuration is a valid ledger state, so is its CPT conjugate. Since the $J$-cost is invariant under all three operations individually (by (i)–(iii)), the combined CPT transformation leaves the action invariant.

This mirrors the Jost–Lüders–Pauli theorem in QFT [8], which derives CPT from Lorentz invariance, locality, and unitarity. In RS, these three conditions are replaced by: ledger conservation (unitarity), voxel locality, and Lorentz invariance of the continuum limit. □

# 6 Weak CP Violation from Generation Torsion

## 6.1 The CKM Phase

Weak CP violation is real and observed: the asymmetry between $B \to J/\psi\, K$ and $\bar{B} \to J/\psi\, \bar{K}$ yields $\sin(2\beta) \approx 0.7$ [4].

**Proposition 6.1** (CKM Phase from Generation Torsion)**.** *In RS, the three fermion generations have integer rungs determined by the generation torsion $\tau_g \in \{0, 11, 17\}$. The CKM matrix arises from the mismatch between up-type and down-type generation structures.*

*Proof.* The rung rule assigns each fermion a position on the $\varphi$-ladder: $r_f = \ell_{\text{sector}} + \tau_g$, where the generation torsion $\tau_g = 0$ (gen 1), $\tau_g = 11$ (gen 2), $\tau_g = 17$ (gen 3). This gives:

| | $\ell_{\text{sector}}$ | Gen 1 | Gen 2 | Gen 3 |
|---|---|---|---|---|
| Leptons | 2 | $r = 2$ | $r = 13$ | $r = 19$ |
| Up quarks | 4 | $r = 4$ | $r = 15$ | $r = 21$ |
| Down quarks | 4 | $r = 4$ | $r = 15$ | $r = 21$ |

The CKM matrix $V = U_u^\dagger U_d$ encodes the rotation between the up-type and down-type mass eigenstates. Although the integer rungs are the same for up and down quarks, the

mass eigenvalues differ because the gap function $\mathrm{gap}(Z) = Z + Z^2 + Z^4$ depends on the charge: $Z_u = 4$ ($Q = +2/3$) and $Z_d = -2$ ($Q = -1/3$).

The CKM mixing angles arise from the 8-tick phase overlaps between different generation states. The complex CKM phase $\delta_{\mathrm{KM}}$ arises because the torsion $\{0, 11, 17\}$ is *asymmetric*: the step from generation 1 to 2 ($\Delta\tau = 11$) differs from the step from 2 to 3 ($\Delta\tau = 6$). This asymmetry, combined with the three independent spatial dimensions, produces an irreducible complex phase. □

## 6.2 Why CP Is Violated in Weak but Not Strong

**Proposition 6.2** (Strong–Weak CP Dichotomy)**.** *CP is preserved in the strong sector and violated in the weak sector, both following from the $J$-cost structure.*

*Proof.* **Strong sector.** The $J$-cost is symmetric under $x \to x^{-1}$ (Theorem 3.1). The QCD action, built from $J$-costs of color-singlet link variables, inherits this symmetry. Therefore $\theta_{\mathrm{QCD}} = 0$ (Theorem 3.2).

**Weak sector.** The electroweak interaction couples to left-handed fermion doublets only: $(u_L, d_L)$, $(\nu_L, e_L)$. This chiral structure explicitly breaks the $x \to x^{-1}$ symmetry for the weak gauge coupling. The $J$-symmetry protects the *gauge* sector from CP violation, but the *Yukawa* sector (which generates masses and mixing) is not constrained to be CP-invariant. The three-generation torsion produces an irreducible complex phase in the CKM matrix. □

## 6.3 The Jarlskog Invariant

The rephasing-invariant measure of CP violation is the Jarlskog invariant [6]:

$$J_{\mathrm{CP}} = \mathrm{Im}(V_{us} V_{cb} V_{ub}^* V_{cs}^*) \approx 3.04 \times 10^{-5}. \tag{11}$$

The smallness of $J_{\mathrm{CP}}$ reflects the CKM hierarchy: off-diagonal elements are suppressed by powers of $\lambda = |V_{us}| \approx 0.225$ (Wolfenstein form), and $J_{\mathrm{CP}} \sim A^2 \lambda^6 \eta$.

*Remark* 6.3. In RS, the Wolfenstein parameter $\lambda$ can be approximated by a $\varphi$-power: $\lambda \approx \varphi^{-3} \approx 0.236$, consistent with the observed $\lambda = 0.2253 \pm 0.0007$ and the Cabibbo angle $\theta_C \approx 13°$. Note: the generation torsion step $\Delta\tau = 11$ suggests $\varphi^{-11/2} \approx 0.071$, which *does not* match $\lambda$. The correct RS identification appears to be $\lambda \approx \varphi^{-3}$ (three lattice spacings), whose structural derivation from the torsion structure is an identified open problem. A complete derivation of all four Wolfenstein parameters from first principles within RS remains an important open problem.

# 7 PMNS CP Phase

**Proposition 7.1** (Leptonic CP Phase — Structural Identification)**.** *The RS structural identification is: $\delta_{\mathrm{CP}} \approx -\pi/2$ ($\approx 270°$), arising from the 8-tick quarter-period phase rotation. This is a structural identification, not a parameter-free derivation; the argument below explains the qualitative origin.*

*Proof.* The 8-tick epoch has a natural quarter-period at $\Delta t = 2$ ticks, corresponding to a phase rotation of $\pi/2$. The PMNS mixing involves transitions between neutrino flavor and mass eigenstates. The quarter-period phase $e^{i\pi/2} = i$ generates a CP phase of $\pi/2$; by convention, this corresponds to $\delta_{\mathrm{CP}} = -\pi/2$ (or equivalently $3\pi/2$). □

The current NuFIT global fit gives $\delta_{\mathrm{CP}} = 197^\circ \pm 40^\circ$ (normal hierarchy) [9], consistent with the RS prediction of $270^\circ$ at the $\sim 2\sigma$ level. DUNE [10] will constrain $\delta_{\mathrm{CP}}$ to $\pm 10^\circ$ by 2030.

# 8 Baryogenesis

Sakharov's three conditions for baryogenesis are [11]:

1. Baryon number violation.
2. C and CP violation.
3. Departure from thermal equilibrium.

In the SM, CKM CP violation is too small to account for the observed baryon asymmetry $\eta_B \sim 6 \times 10^{-10}$ [12].

*Remark* 8.1 (RS Baryogenesis Mechanism). RS provides an additional source of CP violation beyond the CKM matrix: the ledger itself has an intrinsic arrow (the direction of increasing tick count), which breaks T-symmetry at the cosmological level. While CPT is exact (Theorem 5.1), the cosmological boundary condition (the initial ledger state at tick 0) provides the departure from equilibrium and the directional asymmetry needed for baryogenesis. A detailed quantitative derivation of $\eta_B$ from the ledger initial conditions is an open problem.

# 9 Summary: The CP Structure of RS

| **Sector** | **CP Status** | **RS Origin** |
|---|---|---|
| Strong (QCD) | $\theta_{\mathrm{eff}} = 0$ exactly | $J$-symmetry $J(x) = J(x^{-1})$ for bare angle; real $\varphi$-ladder masses for mass phase |
| Weak (CKM) | $\delta_{\mathrm{KM}} \neq 0$ | Asymmetric 3-gen torsion $\{0, 11, 17\}$ with chiral weak coupling |
| Neutrino (PMNS) | $\delta_{\mathrm{CP}} \approx -\pi/2$ | 8-tick quarter-period phase rotation |
| CPT | Exact | Double-entry ledger + voxel locality + Lorentz limit |

# 10 Predictions and Falsifiers

1. **Neutron EDM**: $d_n \lesssim 10^{-32}\ e\cdot\mathrm{cm}$ (SM-only prediction). Detection of $d_n > 10^{-29}\ e\cdot\mathrm{cm}$ would indicate new CP violation beyond the SM and would require RS revision.
2. **No QCD axion**: No axion coupling to $G\tilde{G}$ at any mass. Confirmed detection of a QCD axion would falsify the RS resolution of the strong CP problem.
3. **PMNS CP phase**: $\delta_{\mathrm{CP}} \approx 270^\circ$. DUNE and Hyper-Kamiokande will test this to $\pm 10^\circ$.

4. **CPT invariance**: No CPT violation in any sector. Observation of a particle–antiparticle mass difference (e.g., $m_K \neq m_{\bar{K}}$) would falsify the double-entry ledger.

5. **Cabibbo angle**: $\lambda \approx \varphi^{-3} \approx 0.236$, within 5% of the observed value $0.2253 \pm 0.0007$. (Note: the generation-torsion step $\Delta\tau = 11$ gives $\varphi^{-11/2} \approx 0.071$, which does *not* match $\lambda$; the correct RS identification is $\lambda \approx \varphi^{-3}$, as detailed in the Remark in Section 6.)

# 11 Conclusion

The complete CP structure of physics is derived from RS:

1. $\theta_{\text{eff}} = 0$ from $J(x) = J(x^{-1})$ and the reality of $\varphi$-ladder masses—no PQ axion needed.

2. Weak CP violation from the asymmetric generation torsion $\{0, 11, 17\}$ acting on chiral weak doublets.

3. PMNS CP phase $\delta_{\text{CP}} \approx -\pi/2$ from the 8-tick quarter-period.

4. CPT exact from the double-entry ledger structure.

The strong CP "problem" is dissolved: $\theta_{\text{eff}} = 0$ is a structural identity in a framework where all masses are real and the gauge action is built from the symmetric $J$-cost functional.

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